

Police Strength & Crime Rates:
A Global Analysis

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Introduction

Police are generally known for maintaining order within a given country. While there is a wide range of literature on police, there is currently limited unambiguous explanations of the effects of police strength (Ruddell & Thomas 660).

For this study, police force strength is defined as the number of police per 100,000 population in a given country.

Are levels of crime a function of the number of police or are they more related to broader national political and economic contexts?

There are many questions being asked about police forces' purpose and effectiveness. This study examines the relationship of police force strength and crime conviction rates.

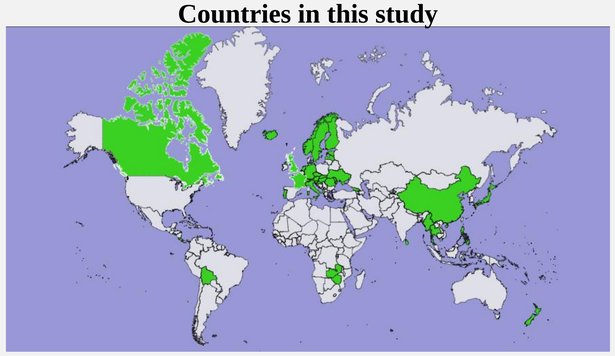
Significance of Study

Police forces around the world are under significant scrutiny. Knowing how police strength affects crime conviction rates can play a role in determining how a country ought to structure its police force.

Methods

A fixed effects time series regression model considers how police strength relates to criminal convictions in 41 countries from 1976 to 2002. The data come from four sources: United Nations Office on Drugs and Crime; the World Bank World Development Indicators; Democracy Time-series Data (Norris 2009); and the Quality of Government Institute.

Countries in this study



	Mean	Std. Dev.	Min	Max
Dependent Variables				
Convictions per 100,000	853	1,217	2.4	8,081
Homicide per 100,000	4.7	5.8	0.2	37.8
Assault per 100,000	94.3	130	1.5	775
Rape per 100,000	6.3	18.5	0.1	204
Theft per 100,000	611	1,349	0.9	7,970
Independent Variables				
Police per 100,000	304	137	87.7	649
% Female in population	51.0	1.1	48.6	54.0
Democracy	0.8	0.4	0	1.0
GDP per capita (in USD)	11,588	11,371	200	43,165
Year			1976	2002

Results

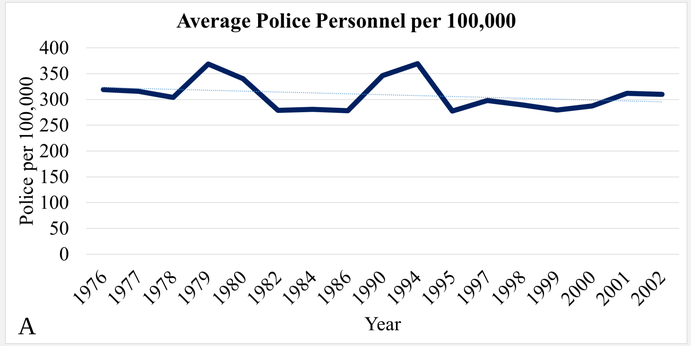


Figure A: reflects the average number of total police personnel per 100,000 people from 1976 to 2002

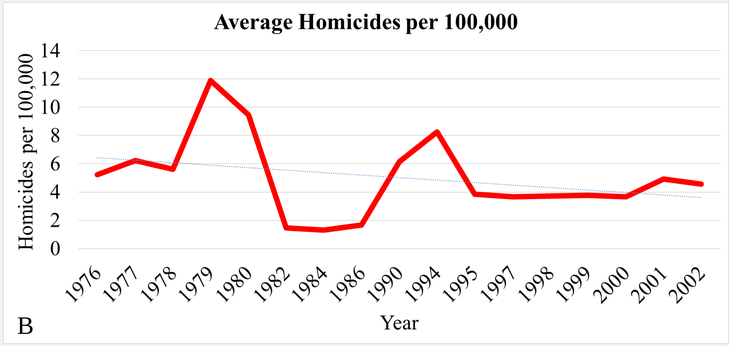


Figure B: reflects the average number of homicide convictions per 100,000 people from 1976 to 2002

Fixed-Effect Regression Results for Five Types of Crime Convictions

	(1) Convictions	(2) Homicide	(3) Theft	(4) Assault	(5) Rape
Police per 100,000	-1.99	0.01*	-0.38	0.23	0.14**
	(1.69)	(0.01)	(1.79)	(0.13)	(0.03)
Population, female (% of total)	-656	-3.77*	-1,244	-99*	1.99
	(595)	(1.86)	(632)	(47.38)	(12.09)
Democracy	1,135*	0.4	847	111**	7.64
	(471)	(1.47)	(500)	(37.49)	(9.57)
GDP per capita current prices (in USD)	-0.04	0	-0.07**	0	0
	(0.02)	(0.00)	(0.02)	(0.00)	(0.00)
Year	61.81**	-0.2**	-76.42**	-7.59**	-0.57
	(22.03)	(0.07)	(23.38)	(1.75)	(0.45)
Constant	-88,779	585**	216,698**	20,081**	984
	(58,308)	(182)	(61,886)	(4,640)	(1,184)
Observations	184	184	184	184	184
Number of Country	41	41	41	41	41
R-squared	0.19	0.11	0.35	0.23	0.13

Standard errors in parentheses
* significant at p<0.05; ** significant at p<0.01

Discussion

Increased police strength is statistically significantly associated with more homicide and rape convictions, but not with other types of convictions.

With police being strongly associated with more homicide convictions, a next step is to incorporate additional independent variables.

Future plans include adding more variables, such as the funding police forces receive, educational data specific to secondary completion and those without schooling, and poverty rates.

To make causal order more clear, there are also plans to lag the independent variables and to interact police strength and democracy.

A key limitation is that not all countries report crime convictions. As a consequence, 136 countries and territories from the original UNODC dataset had to be dropped. The next phase is to develop strategies to reduce missing values and enhance consistency to retain more countries in the analysis.

Lastly, this data doesn't reflect the amount of private and military involvement in policing within countries.

Photo credit: David Mdzinarishvili/Reuters; Andrew Usangie / Photographer's Choice / Getty Images, map created on AMCHARTS.COM

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